MEDICATION SAFETY ASSISTANT

Vision Document

(Small Project)

Version 1.0

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1. Introduction

This Vision Document defines the high-level stakeholder needs, product scope, and capabilities of the Medication Safety Assistant. It explains the problem the product addresses, the people who benefit from it, the intended position of the product, its principal features, and the quality and safety constraints that govern those features.

The product is a responsive Web prototype for safer medication understanding and follow-up. It accepts prescription images, manual entries, typed requests, and voice input. OCR and speech-to-text produce editable text; NLP identifies the user's intent and relevant medication information. Only user-verified data continues to medicine identification and safety assessment. Vietnamese products and active ingredients are identified using medicine-registration data from the Drug Administration of Vietnam (DAV), then mapped to DDInter 2.0 for supported drug–drug interaction checks. The product communicates findings and limitations, distinguishes routine, review-required, and emergency outcomes, and provides immediate escalation guidance for red flags. It does not diagnose, prescribe, or independently change treatment.

This is an initial Inception-phase vision. Detailed use cases, supplementary specifications, implementation decisions, and acceptance tests will refine it during Elaboration and Construction.

1.1 References

No.	Document / source	Date / organization	Location
1	Medication Safety Assistant Project Proposal	October 2026, Group 02	Project repository, PA0 proposal
2	Medication Safety Assistant Software Development Plan, Version 1.0	October 2026, Group 02	Project repository, PA1 Project Plan
3	Medication Without Harm: Policy Brief	7 March 2024, World Health Organization	WHO publication page
4	Publication of drug and medicinal-ingredient registration data in Vietnam	Accessed 16 October 2026, Drug Administration of Vietnam	DAV medicine-registration portal
5	DDInter 2.0: an enhanced drug interaction resource with expanded data coverage, new interaction types, and improved user interface	2025, Tian et al., Nucleic Acids Research, DOI: 10.1093/nar/gkae726	PubMed record
6	Web Content Accessibility Guidelines (WCAG) 2.2	W3C Recommendation, 12 December 2024	W3C WCAG 2.2

2. Positioning

2.1 Problem Statement

Patients may use several medicines at the same time, receive prescriptions from different healthcare professionals, or know medicines only by local brand names. Prescription text may be difficult to enter, and users may not know which active ingredients are involved, whether the available data covers all their medicines, or when a possible interaction needs professional attention. Older adults are a primary user group because they are more likely to manage multiple medicines while also experiencing limited vision, difficulty typing, or unfamiliarity with smartphone applications and online medication resources. These barriers can make it difficult to read medicine labels, search for reliable information, enter prescription details accurately, and understand complex safety warnings without assistance. Missed doses and unclear follow-up further complicate safe medication use for both older adults and their family caregivers.

The problem of	Difficulty capturing, identifying, understanding, and consistently following multiple medicines while recognizing potential medication-safety risks and assessment limitations.
Affects	Primarily older adults, especially those using multiple medicines or having limited vision, digital confidence, or health literacy; family caregivers who assist them; other medicine users; and healthcare professionals who review medication-related questions.
The impact of which is	Users may overlook a potential interaction, misunderstand incomplete results, duplicate an active ingredient, miss a dose, or delay seeking qualified advice. Healthcare professionals may receive incomplete or unstructured information.
A successful solution would	Provide an accessible, plain-language experience for older adults; reduce typing through image and voice input while retaining manual alternatives; require correction and confirmation; collect the safety context needed for assessment; identify Vietnamese products and active ingredients; distinguish routine, review-required, and emergency outcomes without equating unknown with safe; support professional review; and optionally help users follow confirmed schedules.

2.2 Product Position Statement

For	Vietnamese patients, general medicine users, and family caregivers, with a supporting review view for healthcare professionals.
Who	Need a practical way to capture medicines, understand supported safety findings and limitations, organize confirmed schedules, and know when professional review is appropriate.
The Medication Safety Assistant	Is a responsive Web-based medication understanding and safety-support application.
That	Combines image, manual, typed, and voice input with user verification, Vietnamese medicine identification, supported interaction checks, understandable guidance, and medication follow-up.
Unlike	General search engines or chatbots, isolated interaction checkers, and reminder-only applications that address only part of the workflow or may not preserve uncertainty and local product identity.
Our product	Uses DAV data for Vietnamese medicine identification, maps verified active ingredients to DDInter 2.0 for supported interaction checking, keeps AI explanations subordinate to structured findings, immediately escalates emergency red flags, and never treats missing, unresolved, or unreviewed data as proof of safety.

3. Stakeholder and User Descriptions

This section identifies the people and organizations whose needs, constraints, or data affect the product. Direct users are separated from stakeholders who guide, review, or supply information to the prototype.

3.1 Stakeholder Summary

Name / type	Description	Responsibilities and interest

Project team	Three student developers responsible for the course prototype.	Define and prioritize scope; implement NLP, OCR, voice-to-text, data preparation, UI, safety controls, testing, and documentation; disclose limitations and progress.
Course instructors and teaching assistants	Reviewers and evaluators of the software-engineering process and deliverables.	Review scope, planning, evidence, traceability, teamwork, and whether the prototype demonstrates the stated requirements.
Healthcare professionals	Doctors, pharmacists, nurses, or other qualified professionals represented in the review workflow.	Validate the usefulness and clarity of medication summaries; review high-risk or uncertain cases in the prototype; provide professional guidance rather than ceding clinical decisions to the system.
Family caregivers	People who help another person organize and understand medicines.	Represent the needs of users who may have low digital confidence, complex schedules, accessibility needs, or difficulty entering prescription information.
Drug Administration of Vietnam	Government authority whose publication is the primary product-identification source for Vietnamese medicines.	Publishes registration information. It does not endorse this prototype or guarantee that the project's processed copy is current or complete.
DDInter maintainers	Maintainers of the external drug-interaction resource used after active-ingredient mapping.	Provide interaction records, risk levels, mechanisms, recommendations, and coverage information subject to the resource's terms and limitations.

3.2 User Summary

User type	**Description**	**Responsibilities / activities**	**Represented by**
Patient or general user	A person taking one or more medicines who wants help understanding and organizing them.	Upload or capture a prescription, speak or type a request, correct extracted information, provide relevant health context, confirm medicines and schedules, review findings and limitations, optionally enable reminders, and record adherence or possible reactions.	Potential-user interview participants and project team
Family caregiver	A person assisting a relative or dependent with medication management.	Enter and verify medicine information with permission, review schedules and warnings, and help seek professional advice when instructed.	Caregiver interview participants
Healthcare professional reviewer	A qualified professional represented by a simulated review role in the prototype.	Review structured medication summaries, detected risks, unresolved items, possible adverse reactions, and schedule-change requests; request missing information; and record guidance or confirmation.	Healthcare-professional interview participant

3.3 User Environment

The product is used through modern desktop and mobile Web browsers. Patients and caregivers may use it at home, after receiving a prescription, while organizing daily medicines, or before consulting a healthcare professional. They may have only a paper or digital Vietnamese prescription, may know a medicine by brand name, or may prefer to speak rather than type. A camera is needed for prescription capture, a microphone and browser permission are needed for voice input, and notification permission is needed for browser-supported reminders. Manual entry remains available when these capabilities are absent.

Use may occur under time pressure, with background noise, poor lighting, unstable connectivity, small screens, or limited health and digital literacy. Therefore, OCR extractions and voice transcripts must be editable, medical terms must use plain language, warnings must not rely on color alone, and the system must preserve an explicit incomplete/unassessed state. The prototype uses synthetic or de-identified data and is not integrated with hospital, pharmacy, electronic health-record, or real professional-review systems.

Healthcare professional reviewers use a structured Web view, preferably on a desktop or tablet, to inspect verified medicines, source and mapping status, detected risks, stated limitations, symptoms, and review requests.

3.4 Summary of Key Stakeholder or User Needs

Need	Priority	Concerns / cause	Current solution	Proposed solution
Enter medicines with low effort	High	Long prescriptions, spelling, Vietnamese brands, accessibility, or typing difficulty.	Manually copy information or ask another person.	OCR, manual, typed, and voice input with one editable verification flow.
Prevent input errors from becoming safety results	High	OCR and speech recognition may alter names, strengths, or schedules.	Re-read the prescription without systematic validation.	Require correction and explicit confirmation; flag low-confidence or unresolved fields.
Provide enough safety context	High	Interaction significance can depend on the complete medicine list, dose, route, allergies, age, pregnancy, symptoms, and kidney or liver impairment.	Users may omit information or search several sources independently.	Ask for required context, show what is missing, and keep the case incomplete or unassessed until sufficient information is available.
Identify local products and active ingredients	High	Brand names can hide related ingredients or differ from international names.	Search labels or multiple websites manually.	Match verified information to DAV records and retain source provenance.
Understand potential medicine interactions	High	Users may not recognize a risky combination or its significance.	Search the Web or ask a professional when available.	Map ingredients to DDInter 2.0 and present supported interaction findings and severity.
Understand uncertainty and coverage	High	A missing match or absent record may be mistaken for safety.	Results from different sources may be inconsistent or unexplained.	Distinguish detected, no known interaction in available data, incomplete, and unassessed states.

Know when to seek professional advice	High	Users may overreact, ignore a warning, delay emergency care, or change treatment independently.	Contact a doctor or pharmacist without a structured summary.	Separate emergency red flags from review-required cases, provide immediate urgent guidance when needed, and prepare uncertain or unresolved cases for review.
Follow a confirmed medication schedule	Medium	Users forget doses or misread instructions.	Paper notes, alarms, or generic reminder applications.	Offer reminders as an explicit opt-in only after the prescription schedule has been verified and confirmed, then record taken/missed status.
Record possible adverse reactions	Medium	Symptoms may be forgotten or described inconsistently.	Personal notes or memory.	Record symptoms as possible, not confirmed, adverse reactions for follow-up and review.
Protect sensitive health information	High	Images, audio, transcripts, medicine lists, and symptoms are sensitive.	Store information across uncontrolled notes or messaging applications.	Minimize collection and retention, obtain permission, restrict access, and avoid real patient data in the prototype.

3.5 Alternatives and Competition

Alternative	**Strengths**	**Weaknesses**	**Our differentiation**
Direct consultation with a doctor or pharmacist	Qualified clinical judgment and individualized advice.	Access, time, and cost may be limited; users may arrive with incomplete medicine information.	The product does not replace consultation; it prepares a verified, structured summary and signals cases needing review.
General Web search or general-purpose AI chatbot	Convenient and flexible natural-language access.	Source quality and coverage may be unclear; generated responses may imply unsupported certainty.	NLP is limited to input understanding and explanation of structured system findings, with explicit provenance and uncertainty.
Standalone interaction checker	Focused interaction lookup and often broad international drug coverage.	May require users to know normalized ingredient names; may not recognize Vietnamese product names or support intake and follow-up.	DAV-based local product identification precedes DDInter mapping and is connected to verification, guidance, and follow-up.

Medication reminder application	Useful alarms and adherence tracking.	May not verify prescription information or check severe interactions before creating reminders.	Reminders use confirmed schedules and apply a safety gate to affected medicines.
Paper prescription, notes, or phone alarms	Familiar, inexpensive, and usable offline.	Manual transcription, scattered information, limited risk explanation, and weak traceability.	A unified workflow connects capture, verification, safety assessment, guidance, reminders, and follow-up.

4. Product Overview

4.1 Product Perspective

The Medication Safety Assistant is a standalone responsive Web prototype that coordinates external data and processing services. It is not an electronic medical record, a prescribing system, a diagnostic device, or an autonomous clinical decision-maker.

At the product boundary, patients and caregivers provide prescription images, manual text, typed requests, voice input, relevant health information, schedule confirmation, adherence status, and possible adverse-reaction reports. Healthcare professionals interact with a simulated review view. The browser provides camera, microphone, and notification capabilities when permission is granted.

The high-level information flow is:

1. The patient provides medication information and relevant health context through an image, voice, manual entry, or typed request.
2. The system extracts editable fields and marks low-confidence, missing, or conflicting information. Unreliable extraction returns to re-entry or correction rather than continuing to assessment.
3. The patient verifies that the extracted details match the source. This confirmation concerns transcription accuracy, not medical correctness. Any edit invalidates the previous normalized result and starts normalization again.
4. The system performs NLP, DAV identification, ingredient normalization, and reviewed DDInter mapping, then checks whether the required safety context is sufficient. Missing information is requested before reassessment.
5. The system assesses supported risks and chooses one of three routing outcomes: routine/no significant risk detected from available information, professional review required or uncertain, or emergency/red flags.
6. Routine cases receive qualified guidance. Review-required cases remain pending until reviewed; if review is unavailable, the system gives conservative guidance and recommends contacting a qualified doctor or pharmacist. Emergency cases receive immediate urgent guidance without waiting for professional review in the prototype.
7. After reviewing and confirming the final medication plan, the patient may explicitly opt in to reminders. Dose responses and possible adverse reactions are then recorded for follow-up and triage.

OCR transforms printed or digital Vietnamese prescriptions into editable fields. Speech-to-text transforms voice into an editable transcript. NLP identifies one or more intents and medication entities from verified text and may explain structured results, but it does not independently infer safety. DAV publication data supports Vietnamese product and active-ingredient identification. DDInter 2.0 supplies supported interaction information after a reliable mapping is established. External service failure, ambiguous mapping, or insufficient data produces an unavailable, unresolved, incomplete, or unassessed state rather than a safe result. A simulated professional reviewer may approve guidance, request more information, or leave a case pending. An unreviewed case can never be represented as professionally approved.

4.2 Assumptions and Dependencies

Type	Assumption / dependency	Effect if invalid or unavailable
Assumption	Users can read, correct, and explicitly confirm OCR results, voice transcripts, medicines, relevant health context, and schedules.	Unconfirmed information cannot enter safety assessment or create an active reminder.
Dependency	DAV medicine-registration data remains accessible and usable for identifying Vietnamese products and active ingredients.	Identification coverage is reduced; affected products remain unresolved. The system must show the source date and limitations.
Dependency	Prepared DAV-derived normalization data and DAV-to-DDInter mappings can be versioned and reviewed before use.	Unreviewed or ambiguous mappings must be excluded from automatic assessment until corrected.
Dependency	DDInter 2.0 remains accessible and its permitted use supports the course prototype.	Interaction checks may use a permitted controlled snapshot or be unavailable; the interface must disclose coverage and date.
Dependency	OCR can process sufficiently clear printed or digital Vietnamese prescriptions.	Poor-quality, complex, or handwritten content requires manual correction or entry and remains unsupported when unresolved.
Dependency	Speech-to-text supports the user's language, microphone, and browser with acceptable recognition.	Users must receive a clear error and retain typed/manual input; an unconfirmed transcript cannot be assessed.
Dependency	The selected NLP service or model can identify intents/entities and explain only supplied structured facts.	NLP output is withheld or replaced with deterministic text when constraints cannot be enforced or the service is unavailable.
Assumption	Professional review is simulated for the course prototype.	No claim may be made that a real clinician has reviewed a case. Pending or unavailable review remains explicit, while emergency and other appropriate cases direct the user to actual professional care.
Assumption	Modern desktop and mobile browsers provide the required Web, camera, microphone, and notification capabilities.	Unsupported permissions or capabilities require graceful fallback to manual entry and in-application schedules.
Constraint	The team uses synthetic or de-identified development and demonstration data.	Real patient deployment, retention, and clinical integration remain outside the initial scope.

5. Product Features

The following features describe externally visible capabilities of the initial product. They express user value and safety behavior rather than prescribing a specific implementation.

5.1 PF-01: Multimodal Medication and Request Intake

Users can upload or capture a printed/digital Vietnamese prescription, enter medicine information manually, type a request, or speak a request. OCR converts prescription content to editable fields, while speech-to-text converts voice to an editable transcript. The interface shows recording/processing state, permission errors, extraction confidence or uncertainty, and a manual fallback. This feature is high priority.

5.2 PF-02: Verification and Correction

Before medication identification or assessment, users can inspect and correct extracted or transcribed medicine names, strengths, dosage instructions, and schedules. Low-confidence, missing, conflicting, and unresolved fields are visibly identified. Explicit confirmation creates the verified input used by downstream functions. The user confirms that the extracted details match the source, not that they are medically correct. Any later edit invalidates the previous normalization, mapping, and assessment and causes them to be run again. This feature is high priority and safety-critical.

5.3 PF-03: Intent Identification and Conversational Assistance

The system identifies one or more supported intents from typed or confirmed transcribed requests, such as checking interactions, explaining a structured finding, managing a reminder, or recording a possible reaction. It requests clarification when the intent or medicine is ambiguous. Conversational output may restate structured findings in plain language but cannot create an independent safety conclusion. This feature is high priority.

5.4 PF-04: Vietnamese Medicine Identification and Normalization

The system matches verified Vietnamese medicine information to prepared DAV records, identifies active ingredients and relevant registration attributes, and preserves source and version information. Ambiguous or unmatched products are clearly unresolved. Prepared data and active-ingredient mappings are versioned and reviewed before use. This feature is high priority and safety-critical.

5.5 PF-05: Medication Safety and Interaction Assessment

Reliably identified active ingredients are mapped to reviewed DDInter 2.0 entities. The system checks supported drug–drug interactions and severity. A result has an explicit state: interaction detected, no known interaction found within the available data, incomplete, unassessed, or temporarily unavailable. Before assessment, the system checks for required context such as the complete medicine list, dose, route, allergies, age, pregnancy, symptoms, and relevant kidney or liver impairment. It requests missing information instead of assuming that an unknown condition is safe. The resulting route is routine/no significant risk detected from available information, professional review required or uncertain, or emergency/red flags. Therapeutic duplication and drug–disease checks are included only when the team can establish adequate data, mapping, and acceptance tests. This feature is high priority and safety-critical.

5.6 PF-06: Safety Findings and Explainable Guidance

Users receive the involved medicines, severity where available, a concise explanation grounded in the structured interaction record, source/coverage limitations, and an appropriate next step. Warnings never instruct users to start, stop, replace, or change a prescribed medicine independently. Severe, uncertain, unresolved, or concerning cases advise qualified professional review. Emergency red flags receive immediate instructions to seek urgent real-world help and never wait for the simulated review workflow. This feature is high priority and safety-critical.

5.7 PF-07: Professional Review Support

The prototype prepares a structured case summary containing verified medicines, active ingredients, mapping status, detected findings, limitations, schedules, and reported symptoms. A simulated professional reviewer can record

guidance or schedule confirmation, or request more information from the patient before the case is reassessed. A case without a response remains pending or unavailable; the system provides conservative interim guidance and recommends an external doctor or pharmacist rather than implying approval. The product labels this workflow as a simulation and does not imply that real clinical review has occurred. This feature is high priority for demonstrating the safety workflow.

5.8 PF-08: Confirmed-Schedule Medication Reminders

After reviewing and confirming the medication plan, users can separately opt in to reminders based on verified prescription instructions and choose notification times that remain within those instructions. Plan confirmation does not automatically activate reminders. Changes to the prescribed dosing schedule require professional confirmation and user acceptance. A new reminder for a medicine involved in a detected severe interaction remains on hold until review and confirmation; unrelated reminders may proceed. This feature is medium priority.

5.9 PF-09: Adherence and Possible Adverse-Reaction Follow-up

Users can respond to a reminder by recording a dose as taken, missed, or skipped, and the system stores the resulting adherence event. Users can also record symptoms for follow-up. Symptoms are labeled as possible adverse drug reactions, not proven causation, and are triaged into mild/self-monitoring, professional review required, or emergency/red flags. Review and emergency outcomes follow their respective routes. This feature is medium priority.

5.10 PF-10: Medication and Assessment History

Users can review their verified medicine list, prior assessments, result states, reminders, adherence events, possible-reaction records, and review guidance. History preserves the relevant data-source/version context so that a previous result is not presented as a current guarantee. This feature is medium priority and may be reduced if the prototype cannot provide adequate privacy controls.

6. Non-Functional Requirements

The following requirements apply across the product. “Must” identifies an acceptance condition for the initial prototype; targets may be refined after technical experiments, but changes require review because several conditions are safety-critical.

ID	Category	Measurable requirement	Priority	Source
NFR-01	Medical safety	The system must never independently prescribe, diagnose, or instruct a user to start, stop, replace, or change the dose or schedule of a prescribed medicine. Safety-critical acceptance tests must find no prohibited instruction in supported result flows.	High	Proposal / interviews
NFR-02	Input safety	Every OCR extraction and voice transcript used for identification or assessment must have an explicit user-confirmed version. Tests must verify that unconfirmed input cannot start an assessment and that editing confirmed medication data invalidates and recomputes normalization, mapping, and assessment.	High	Product constraint

NFR-03	Uncertainty	Every assessment must display an explicit data state (detected interaction, no known interaction in available data, incomplete, unassessed, or unavailable) and a routing outcome (routine, review required/uncertain, or emergency). Missing or unresolved data must never be rendered as safe.	High	Product constraint
NFR-04	AI safety	NLP-generated explanations must be grounded only in structured system findings and approved guidance fields. Unsupported medication conclusions must be rejected or replaced with deterministic safe text in the test suite.	High	Project scope
NFR-05	Mapping integrity	Each assessed medicine must link to its verified input, DAV record/version, normalized ingredient, reviewed DDInter mapping, and interaction-source version. Ambiguous mappings must remain unresolved.	High	Data workflow
NFR-06	Reminder safety	Reminder activation must require a confirmed medication plan and a separate explicit user opt-in. A new reminder for a medicine involved in a detected severe interaction must remain inactive until the recorded professional-review and user-confirmation conditions are satisfied.	High	Proposal
NFR-07	Privacy	The prototype must request microphone/camera permission at the point of use, collect only data required for the selected function, and avoid retaining raw audio after transcription unless the user explicitly agrees and retention is necessary.	High	User need
NFR-08	Test-data privacy	Source control, screenshots, logs, demonstrations, and shared evidence must contain only synthetic or de-identified health data; repository review must find no API secret or real patient record.	High	Project constraint
NFR-09	Access control	Professional-review information and user history must not be accessible through another user's ordinary session. Authorization tests must cover direct URL/API access to protected records.	High	Stakeholder need
NFR-10	Failure behavior	If OCR, speech-to-text, NLP, DAV data, or DDInter is unavailable, the system must identify the failed step, preserve entered data when safe, provide a retry or manual fallback, and avoid presenting a completed assessment.	High	Reliability need
NFR-11	Performance	Under the course test environment, 95% of ordinary page actions excluding external OCR, speech, NLP, and interaction-service latency should respond within two seconds; external processing must show progress within one second.	Medium	User need
NFR-12	Responsive use	All primary patient flows must be usable without horizontal page scrolling at representative widths of 360, 768, and 1280 pixels, except intrinsically wide data visualizations.	Medium	Product scope

NFR-13	Accessibility	Primary flows should conform to applicable WCAG 2.2 Level AA success criteria. All functions must have a keyboard-accessible path; warnings must not rely on color alone; voice and camera input must have manual alternatives.	Medium	W3C WCAG 2.2
NFR-14	Understandability	In usability review, at least 4 of 5 representative participants must correctly distinguish “no known interaction in available data” from “safe” and identify the recommended next step for a severe or unresolved case.	High	Interview objective
NFR-15	Compatibility	The release candidate must pass the primary intake-to-guidance flow on the latest stable Chrome and Edge desktop versions and on at least one Android mobile browser available to the team.	Medium	User environment
NFR-16	Auditability	Safety-critical processing must record time, verified input version, source/mapping versions, result state, and review/reminder decision without storing raw secrets or unnecessary sensitive content in logs.	High	Safety review
NFR-17	Maintainability	NLP, OCR, speech-to-text, DAV preparation/mapping, DDInter assessment, reminders, and presentation must have documented boundaries so one external service can be replaced without rewriting unrelated product flows.	Medium	Project team
NFR-18	Documentation	The prototype must include a concise user guide covering supported input, verification, result states, limitations, privacy choices, professional escalation, reminder rules, and known unsupported cases.	Medium	Stakeholders
NFR-19	Context completeness	Before assessment, the system must validate the required medication and health-context fields. Missing required context must produce a correction/request state and must not produce a routine or safe conclusion.	High	Safety workflow
NFR-20	Escalation behavior	Emergency/red-flag outcomes must immediately display urgent real-world guidance without waiting for review. Review-required cases without a response must remain pending or unavailable and must never be automatically approved.	High	Safety workflow
NFR-21	ADR triage	Each possible adverse-reaction report must receive exactly one routing outcome: mild/self-monitoring, professional review required, or emergency/red flags. Emergency routing must satisfy NFR-20.	High	Follow-up workflow